This assignment will not be graded and is only for practice.

Level 1

1) Choose the correct statements about $f(x, y) = x^3 + xy + y^4 - 4$. *1 point*

- [ ] $(0, 0)$ is a critical point of f.
- [ ] $(0, 0)$ is the only critical point of f in R^2.
- [ ] $f_{xx} = 6x$
- [ ] $f_{yy} = 12x^2$
- [ ] $f_{xy} = 1$
- [ ] $f_{yx} = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f.
$f_{xx} = 6x$
$f_{xy} = 1$

2) If $f(x, y) = \tan^{-1}(x/y)$, then which of the following options are correct. *1 point*

- [ ] $f_x = \frac{y}{(x^2+y^2)}$
- [ ] $f_{yy} = \frac{2xy}{(x^2+y^2)^2}$
- [ ] $f_{yy} = \frac{-2xy}{(x^2+y^2)^2}$
- [ ] $f_{yx} = \frac{x^2-y^2}{(x^2+y^2)^2}$
- [ ] $f_{yx} = \frac{y^2-x^2}{(x^2+y^2)^2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x = \frac{y}{(x^2+y^2)}$
$f_{yy} = \frac{2xy}{(x^2+y^2)^2}$
$f_{yx} = \frac{x^2-y^2}{(x^2+y^2)^2}$

3) If $f(x, y) = x^2 - 3xy^3$, then the determinant of the Hessian matrix $Hf(x, y)$ is **1 point**

- $36xy + 81y^4$
- 0
- $-36xy - 81y^4$
- $36xy - 81y^4$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$-36xy - 81y^4$

Level 2

4) Choose the correct statements about $f(x, y) = x^2 + y^4$. **1 point**

- $(0, 0)$ is a critical point of f.
- Rank of the Hessian matrix $Hf(0, 0)$ is 2.
- The determinant of the Hessian matrix $Hf(0, 0)$ is 0.
- The determinant of the Hessian matrix $Hf(0, 0)$ is 1.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f.
The determinant of the Hessian matrix $Hf(0, 0)$ is 0.

5) Number of critical points of the function $f(x, y) = x^3 + y^3 - 3x - 3y$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

6) For which of these functions are the mixed partial derivatives the same at $(0, 0)$.

1 point

- [ ] $f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$.
- [ ] $f(x, y) = \{\begin{matrix} 1 & y = 0 \\ 0 & y = 0. \end{matrix}$
- [ ] $f(x, y) = |x|(y^2 + 1)$.
- [ ] $f(x, y) = ax + by$, where a and b are two fixed real numbers.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$.
$f(x, y) = ax + by$, where a and b are two fixed real numbers.

Your score is: 0/6